

# Christopher Aytona

Data and Reporting Analyst | Automation & Data Engineering

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## SUMMARY

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Data and automation engineer building production-grade ETL pipelines, real-time monitoring systems, and AI-driven operational tools at enterprise scale. 8+ years of software engineering across startups, operations technology, and applied AI — including published research on autonomous agent architecture. Systems currently serve 800+ daily users across 35+ facilities. Key results: reduced ticket resolution from 36 hours to under 1 hour, eliminated 40+ hours/week of manual reporting across 8 sites, supported site Quality KPI to #1 network ranking.

## TECHNICAL SKILLS

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**Languages:** Python, SQL, JavaScript/TypeScript, VBA, C#

**Data:** Amazon Redshift (Spectrum), DynamoDB, MySQL, S3 data lakes, pandas, ETL/ELT design

**Cloud:** AWS (Lambda, EventBridge, Step Functions, ECS Fargate, CDK, CloudWatch, Bedrock, API Gateway)

**Automation:** Slack API, Selenium, Playwright, Flask, boto3, CI/CD pipelines

**Visualization:** QuickSight (Author), Excel/VBA, Chart.js, HTML dashboards

**Methods:** Event-driven architecture, Agile, Lean/Six Sigma, infrastructure-as-code

## PROFESSIONAL EXPERIENCE

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### Amazon

Sep 2021 – Present

Belleville & Ajax, ON — AMXL (Extra Large) Fulfillment Network

**Data and Reporting Analyst (L4)** — Sep 2023 – Present    **Process Assistant (L3)** — Jan 2023 – Sep 2023

**Fulfillment Associate (L1)** — Sep 2021 – Jan 2023

- Architected 36+ automated ETL pipelines (Python, Redshift Spectrum, Lambda + EventBridge) delivering 350+ scheduled outputs per week across 35+ sites and 800+ daily users — zero manual intervention after deployment
- Designed event-driven compliance monitoring platform (17 microservices, 8 sites) in partnership with regional operations leadership — proactively flags labour law violations before breach across 12hr/day and 60hr/week jurisdictions
- Built cross-site Slack integration platform (13 webhook integrations, 6 FCs) providing real-time backlog alerts, safety rotation tracking, and capacity notifications to regional leadership — replaced manual email-based reporting
- Engineered AI ticket triage system (AWS Bedrock Claude, DynamoDB, Lambda) that classifies, routes, and auto-resolves routine tickets — cut mean resolution from 36 hours to under 1 hour while surfacing miscategorized high-impact issues to site leadership
- Designed inbound high-risk ASIN flagging system (IHRA) scoped across 32 AMXL FCs network-wide — identifies cross-contamination risk patterns in stow operations before inventory damage occurs
- Built autonomous agent orchestration layer (11 scheduled services, persistent memory, self-healing) reducing operational automation costs by 70% while managing monitoring, reporting, and incident response across all systems
- Conducted 170-row inventory defect deep dive using Redshift warehouse data — traced root causes through bin history, transship records, and cycle count evidence chains, identifying 12 distinct operator error patterns

- Partnered with Quality, ICQA, and Transportation teams to deliver weekly KPI bridges, defect root-cause analyses, and data-driven process recommendations — supported site to #1 network Quality ranking and sustained Top 5
- Owned site Quality KPI performance as subject matter expert — provided data analysis and reporting that informed process changes adopted at regional level
- Promoted L1→L3 in 16 months; managed shift operations for 30+ associates across inbound, stow, and ICQA departments with cross-functional coordination

## CEO & Founder

Jun 2025 – Present

TapOrbit Studios — Toronto, ON (Side Venture)

- Founded indie game studio — own day-to-day operations, finances, team management, and project timelines
- Shipped debut title *Zombie Citizen* to iOS and Android (2026); managed full release cycle from development through store submission and post-launch support
- Contribute directly to Unity/C# codebase when needed to unblock technical work or prototype features

## Software Developer

Sep 2018 – Oct 2019

Treasured — Markham, ON

- R&D on VR/AR virtual world SaaS product (Oculus, Magic Leap); partnered with CTO on architecture and technical roadmap
- Prototyped spatial interaction systems and contributed to core rendering pipeline

## Game Developer / Engineering Lead

2014 – 2018

Transhumanoid Productions (Remote) & Ruckus Games (Toronto)

- Led distributed engineering teams of 4–8 across multiple time zones; owned full development lifecycles from concept to ship
- Shipped 5 products end-to-end in Unity/C# — gameplay systems, networking, UI frameworks, and build pipelines
- Progressed from Developer to Team Lead (Transhumanoid) and Engineering Lead (Ruckus) based on delivery track record

## PUBLICATIONS

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Aytona, C. (2026). "SHARD: Self-Healing Autonomous Resilient Delegation — A Pattern Language for Persistent AI Agent Harnesses." Preprint, Zenodo/SSRN. DOI: 10.5281/zenodo.20474819

Aytona, C. (2026). "Resource-Efficient Architectures for Persistent Agent Systems." In preparation (target: AAAI 2027).

## EDUCATION

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**Bachelor of Computer Science** — Ontario Tech University, 2021

**Advanced Diploma in Game Programming** — George Brown Polytechnic, 2017 — Graduate with Honours